Practice test AAHL MARKSCHEME (Funtions) Name -------- [70 marks]

1. [Maximum mark: 9]
Let $f(x) = \frac{2x^2-5x-12}{x+2}$, $x \in \mathbb{R}$, $x \neq -2$.

(a) Find all the intercepts of the graph of $f(x)$ with both the x and y axes. [4]

Markscheme

$x = 0 \Rightarrow y = -6$ intercept on the y axes is (0, –6) ***A1***

$2x^2 - 5x - 12 = 0 \Rightarrow (2x+3)(x-4) = 0 \Rightarrow x = \frac{-3}{2}$ or 4 ***M1***

intercepts on the x axes are $\left(\frac{-3}{2}, 0\right)$ and $(4, 0)$ ***A1A1***

[4 marks]

(b) Write down the equation of the vertical asymptote. [1]

Markscheme

$x = -2$ ***A1***

[1 mark]

(c) As $x \to \pm\infty$ the graph of $f(x)$ approaches an oblique straight line asymptote.

Divide $2x^2 - 5x - 12$ by $x + 2$ to find the equation of this asymptote. [4]

Markscheme

$f(x) = 2x - 9 + \frac{6}{x+2}$ ***M1A1***

So equation of asymptote is $y = 2x - 9$ ***M1A1***

[4 marks]

2. [Maximum mark: 12]
The function f is defined by $f(x) = 5(x+1)(x+3)$, where $x \in \mathbb{R}$.

(a) Write $f(x)$ in the form $a\left(x - h^2\right) + k$, where $a, h, k \in \mathbb{Z}$. [4]

Markscheme

METHOD 1

$a = 5$ ***(A1)***

attempt to use roots and symmetry to find h ***(M1)***

$h = \frac{(-1)+(-3)}{2}$ **OR** half the distance between the roots $\frac{(-1)-(-3)}{2} = 1$ (may be seen on a diagram)

$h = -2$ (accept $x = -2$) *(A1)*

$f(x) = 5(x - (-2))^2 - 5\left(= 5(x+2)^2 - 5\right)$ *A1*

$(a = 5,\ h = -2,\ k = -5)$

METHOD 2

$a = 5$ *(A1)*

attempt to expand

$(x+1)(x+3) = x^2 + 4x + 3$ **OR** $5(x+1)(x+3) = 5x^2 + 20x + 15$

EITHER

uses their expansion to attempt to complete the square to the form *(M1)*

$p(x+q)^2 + r$, where q is half the coefficient of their x term

$= (x+2)^2 - 2^2 + 3\left(= (x+2)^2 - 1\right)$ **OR** $5\left[(x+2)^2 - 2^2 + 3\right]\left(= 5(x+2)^2 - 5\right)$ *(A1)*

OR

uses their expansion to attempt to differentiate and sets equal to zero *(M1)*

$\frac{\mathrm{d}y}{\mathrm{d}x} = 2x + 4 = 0$ **OR** $\frac{\mathrm{d}y}{\mathrm{d}x} = 10x + 20 = 0$

$h = -2$ (accept $x = -2$) *(A1)*

OR

uses their expansion to attempt to find axis of symmetry using $h = \frac{-b}{2a}$ *(M1)*

$h = \frac{-4}{2}$ **OR** $h = \frac{-20}{10}$

$h = -2$ (accept $x = -2$) *(A1)*

THEN

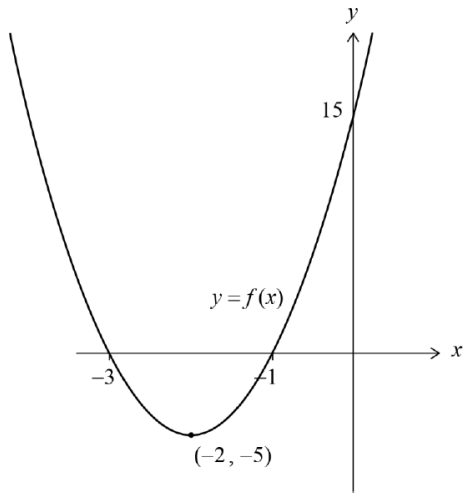
$f(x) = 5(x - (-2))^2 - 5\left(= 5(x+2)^2 - 5\right)$ *A1*

$(a = 5,\ h = -2,\ k = -5)$

[4 marks]

(b) Sketch the graph of $y = f(x)$, showing the values of any intercepts with the axes and the coordinates of the vertex. [4]

Markscheme



M1A1A1A1

award ***M1*** for a roughly symmetric curve which is concave up

award ***A1*** for x intercepts at -3 and -1

award ***A1*** for y intercept at 15

award ***A1*** for vertex at $(-2, -5)$

[4 marks]

(c) Solve the inequality $f(x) \leq 40$. [4]

Markscheme

$5(x+2)^2 - 5 \leq 40$ **OR** $5(x+1)(x+3) \leq 40$ **OR** $(x+1)(x+3) \leq 8$ leading to $(x+2)^2 \leq 9$ **OR** $5x^2 + 20x - 25 \leq 0$ **OR** $x^2 + 4x - 5 \leq 0$ ***(A1)***

valid attempt to find the critical values for their quadratic inequality ***(M1)***

$x + 2 = \pm 3$ **OR** $(x+5)(x-1) = 0$

$x = -5, x = 1$ ***(A1)***

$-5 \leq x \leq 1$ ***A1***

Note: Accept $(x \in)[-5, 1]$ or equivalent.

[4 marks]

3. [Maximum mark: 7]

The quadratic equation $x^2 + kx + 15 - k = 0$ has two distinct real roots.

(a) Find the possible values of k. [5]

Markscheme

recognising $\Delta > 0$ (seen anywhere) *(M1)*

$\Delta = k^2 - 4(15 - k)$ $\left(= k^2 + 4k - 60\right)$ *A1*

valid attempt to solve quadratic (in)equality *(M1)*

$(k-6)(k+10)$ **OR** $k = \frac{-4 \pm \sqrt{4^2 - 4(-60)}}{2}$

two correct values -10 and 6 (seen anywhere) *A1*

$k < -10,\ k > 6$ *A1*

[5 marks]

(b) Find the possible values of k in the case where the two distinct real roots are both positive or both negative. [2]

Markscheme

recognising constant or product of roots is positive *(M1)*

$15 - k > 0 \Rightarrow k < 15$

$k < -10,\ 6 < k < 15$ *A1*

[2 marks]

4. [Maximum mark: 5]

Consider the function $h(x) = \log_{10}\left(3x^2 - rx + r - 2\right)$, where $x \in \mathbb{R}$.

Find the possible values of r. [5]

Markscheme

METHOD 1

recognition that $3x^2 - rx + r - 2$ must be greater than zero (seen anywhere) *R1*

(discriminant $=$) $(-r)^2 - 4(3)(r-2)$ $\left(= r^2 - 12r + 24\right)$ (seen anywhere) *(A1)*

$2.53589\ldots$ $\left(= 6 - 2\sqrt{3}\right)$ AND $9.46410\ldots$ $\left(= 6 + 2\sqrt{3}\right)$ (seen anywhere) *(A1)*

recognition that discriminant of $3x^2 - rx + r - 2$ is less than zero *(M1)*

$2.54 < r < 9.46$ $\left(6 - 2\sqrt{3} < r < 6 + 2\sqrt{3}\right)$ *A1*

Note: Accept $2.54 \le r \le 9.46$.

METHOD 2

recognition that $3x^2 - rx + r - 2$ must be greater than zero (seen anywhere) ***R1***

EITHER

minimum when $x = \frac{r}{6} \Rightarrow \left(y =\right) 3\left(\frac{r}{6}\right)^2 - r\left(\frac{r}{6}\right) + r - 2 \ \left(> 0\right)$ ***(A1)***

attempt to solve their inequality for y (must be in terms of r and r^2) ***(M1)***

OR

$x < 1 \Rightarrow r > \frac{3x^2-2}{x-1}$ OR $x > 1 \Rightarrow r < \frac{3x^2-2}{x-1}$ ***(A1)***

attempt to find local minumum AND local maximum of $r = \frac{3x^2-2}{x-1}$ ***(M1)***

THEN

$\left(r >\right) 2.53589\ldots \ \left(= 6 - 2\sqrt{3}\right)$ AND $\left(r <\right) 9.46410\ldots \ \left(= 6 + 2\sqrt{3}\right)$ (seen anywhere) ***(A1)***

$2.54 < r < 9.46 \ \left(6 - 2\sqrt{3} < r < 6 + 2\sqrt{3}\right)$ ***A1***

Note: Accept $2.54 \leq r \leq 9.46$.

[5 marks]

5. [Maximum mark: 5]

(a) Solve $3m^2 + 5m - 2 = 0$. [3]

Markscheme

valid attempt to solve a quadratic equation (factorising, use of formula, completing square, or otherwise) ***(M1)***

$(3m - 1)(m + 2) = 0$ OR $m = \frac{-5 \pm \sqrt{25+24}}{6}$ (or equivalent) ***(A1)***

$m = \frac{1}{3},\ m = -2$ ***A1***

[3 marks]

(b) Hence or otherwise, solve $3 \times 9^x + 5 \times 3^x - 2 = 0$. [2]

Markscheme

setting their m-value(s) $= 3^x$ OR recognising a quadratic in 3^x ***(M1)***

$3^x = \frac{1}{3}$ (or $3^x = -2$) OR $3 \times \left(3^x\right)^2 + 5 \times 3^x - 2 = 0$

$x = -1$ ***A1***

Note: Award the final ***A1*** if candidate's answer includes $x = -1$ and $x = \log_3(-2)$. Award ***A0*** if other incorrect answers are given.

[2 marks]

6. [Maximum mark: 5]

Given that $Ax^3 + Bx^2 + x + 6$ is exactly divisible by $(x + 1)(x - 2)$, find the value of A and the value of B. [5]

Markscheme

A=1 & B=-4

7. [Maximum mark: 6]

f(x) = $\frac{4-x^2}{4-\sqrt{x}}$

a) State the largest possible domain of f(x) [2]

b) Solve the inequality $f(x) \geq 1$ [4] [6]

Markscheme

$a)\ x \geq 0\ and\ x \neq 16$ $[2] b)\ 0 \leq x \leq 1 \leq\ and\ x > 16\ \ by\ graph\ \ [4]$

8. [Maximum mark: 5]

The quadratic equation $(k-1)x^2 + 2x + (2k-3) = 0$, where $k \in \mathbb{R}$, has real distinct roots.

Find the range of possible values for k. (Use of calculator) [5]

Markscheme

* This sample question was produced by experienced DP mathematics senior examiners to aid teachers in preparing for external assessment in the new MAA course. There may be minor differences in formatting compared to formal exam papers.

attempts to find an expression for the discriminant, Δ, in terms of k **(M1)**

$\Delta = 4 - 4(k-1)(2k-3)\ \ \left(= -8k^2 + 20k - 8\right)$ **(A1)**

Note: Award **M1A1** for finding $x = \frac{-2 \pm \sqrt{4-4(k-1)(2k-3)}}{2(k-1)}$.

attempts to solve $\Delta > 0$ for k **(M1)**

Note: Award **M1** for attempting to solve $\Delta = 0$ for k.

$\frac{1}{2} < k < 2$ **A1A1**

Note: Award **A1** for obtaining critical values $k = \frac{1}{2}, 2$ and **A1** for correct inequality signs.

[5 marks]

9. [Maximum mark: 7]

Consider the function $f(x) = 7\sqrt{x} - x - 7$, where $0 \leq x \leq 60$.

(a) Find the value of

(a.i) $f(0)$; [1]

Markscheme

$f(0) = -7$ *A1*

[1 mark]

(a.ii) $f(60)$. [1]

Markscheme

$-12.7782\ldots$

$f\left(60\right) = -12.8 \quad \left(= 7\sqrt{60} - 67 = 14\sqrt{15} - 67\right)$ *A1*

[1 mark]

(b) Find the two roots of $f(x) = 0$. [2]

Markscheme

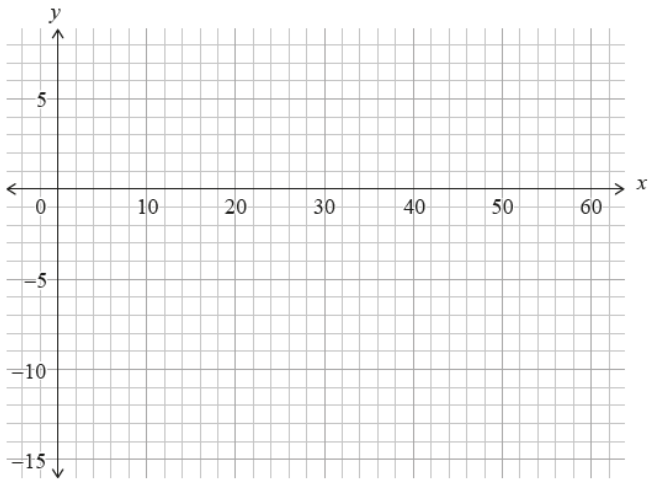
attempt to find at least one root *(M1)*

$x = 1.46098\ldots$ and $x = 33.5390\ldots$

$x = 1.46$ and $x = 33.5$ *A1*

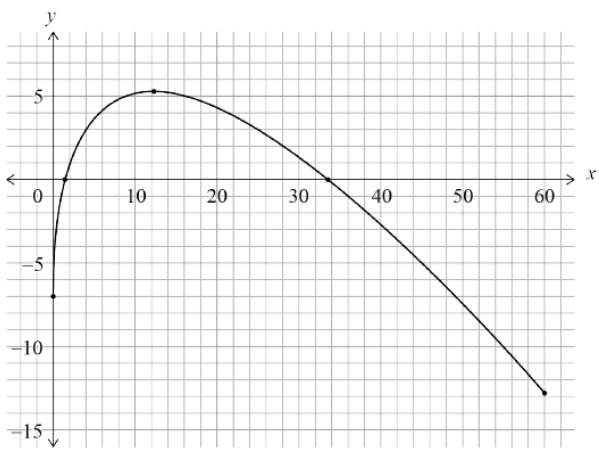
[2 marks]

(c) Sketch the graph of $y = f(x)$ on the following grid.



[3]

Markscheme



A1A1A1

Note: Follow through from their part (a).

Award ***A1*** for endpoints at approximately $(0,\ 7)$ and $(60,\ -12.8)$. Allow for y-intercept $-8 < y < -6$ and for the right endpoint in the interval $x = 60,\ -13 < y < -12$.

The following two ***A*** marks may only be awarded if the approximate shape is correct.

Award ***A1*** for x-intercepts at approximately $x = 1.5$ and $x = 33.5$ and award ***A1*** for maximum at approximately $(12.2,\ 5.3)$. Allow for x-intercepts in the intervals $0 < x < 2,\ 32 < x < 34$, and maximum in the intervals $10 < x < 14,\ 4.5 < y < 6$.

[3 marks]

10. [Maximum mark: 5]

Solve the inequality $|2x - 3| \geq |x + 1|$. [5]

Markscheme

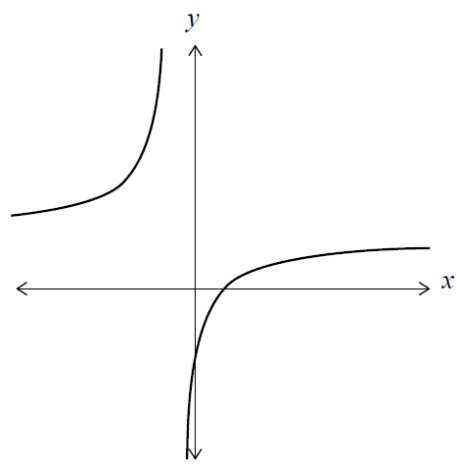
use graph or sign diagram

$(-\infty < x \leq .667 \ \ and \ 4 \leq x < \infty$

11. [Maximum mark: 4]

The function f is defined by $f(x) = \frac{3x-2}{2x+1}$ for $x \in \mathbb{R}, x \neq -\frac{1}{2}$.

The following diagram shows part of the graph of $y = f(x)$.



(a) Write down the value of $f(0)$. [1]

Markscheme

-2 (accept $(0, \ -2)$) ***A1***

[1 mark]

(b) Write down the equation of the horizontal asymptote. [1]

Markscheme

$y = \frac{3}{2}$ (must be an equation) ***A1***

Note: Do not accept $\neq$ sign.

[1 mark]

The function g is defined by $g(x) = -f(x)$ for $x \geq 0$.

(c) Find the range of g. [2]

Markscheme

attempt to reflect the graph of f in the x-axis **OR** one correct value seen *(M1)*

both correct values $-\frac{3}{2}$ and 2 (seen anywhere) *A1*

$-\frac{3}{2} < y \leq 2$ *A1*

[3 marks]

© International Baccalaureate Organization, 2026